Strategic Plan to Recover the Performance of a Mining Equipment Fleet

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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June 2021
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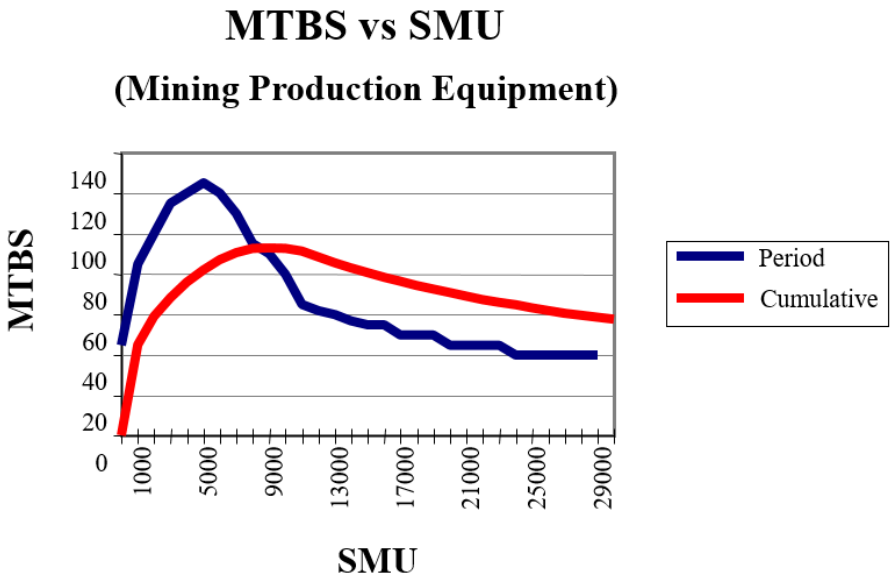
1.0 Introduction

Maintenance and repair programs dedicated to support and ensure mining equipment performance are always challenged by application type and production plan requirements. Mining customers continuously seek operational excellence to achieve expected production at the optimum cost/ton.

In this scenario, equipment performance is a key variable to control and maintain over time, motivating production and maintenance departments to work together toward common goals, of which Safety, Availability, Reliability, Productivity, and Cost per Ton are the usual goals considered. The maintenance department leads control for availability and reliability while the production department is responsible for utilization, productivity and operating the equipment within specifications. Both departments share the responsibility for safety and cost per ton.

Ideally, joint efforts of the production and maintenance departments, supported by the equipment supplier, should ensure proper and stable equipment performance results. However, due to several internal and/or external factors, the equipment performance can be affected and start to decline over time. The first indication of this undesired situation is normally reflected in equipment availability and reliability. Subsequently, this will negatively affect production results.



Typical behavior of mining production equipment in terms of MTBS vs SMU (Hours)

This normal behavior, as illustrated in the graph above, can be aggravated if the maintenance effort is not increased, affecting availability and reliability results. When maintenance departments face this type of challenge, there are several improvement options to consider, namely improving maintenance processes or machine application conditions. Improving M&R processes is a “long term” effort that requires intensive discipline and focus of the maintenance organization, which at the same time, is being challenged by frequent unscheduled events.

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This best practice describes the strategic foundational elements and methodology to improve equipment fleet availability and reliability in a relatively "short time". This approach is based on the concept to identify and recover/repair the worst performing equipment.

There are multiple advantages of applying this option, the most important being:

- Improved fleet performance results will be achieved in a shorter time.
- Each repaired machine will reduce the amount of support required.
- Program results can be measured machine-by-machine, rather than the whole fleet.
- Repair programs can be executed by a separate parallel crew, without disturbing the regular maintenance operation.

This best practice is the result of strategies, concepts, and experience gained when applied to a fleet of (61) Caterpillar 797F OHT operating in Buenavista del Cobre (BVC), Cananea, Sonora, Mexico. The owner of the BVC operation, Grupo Mexico, agreed to join efforts with MATCO and Caterpillar to implement and execute the 797F fleet recovery.

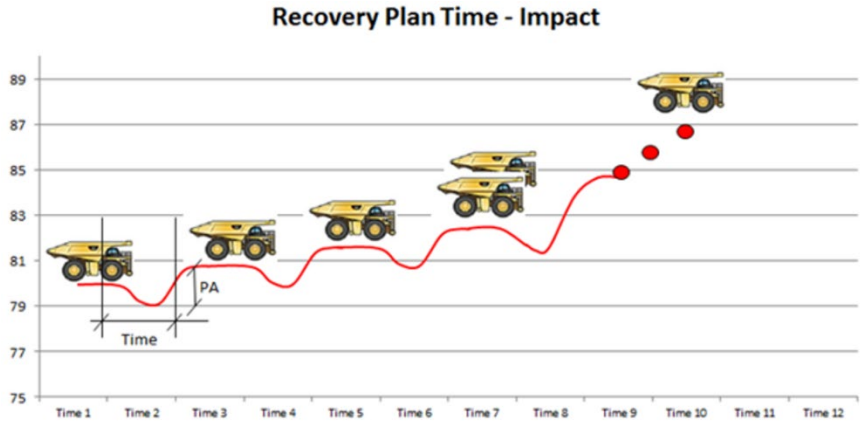
Note: Although this BP was applied to 797F trucks, the concept is applicable to other mobile fleets.

2.0 Best Practice Description

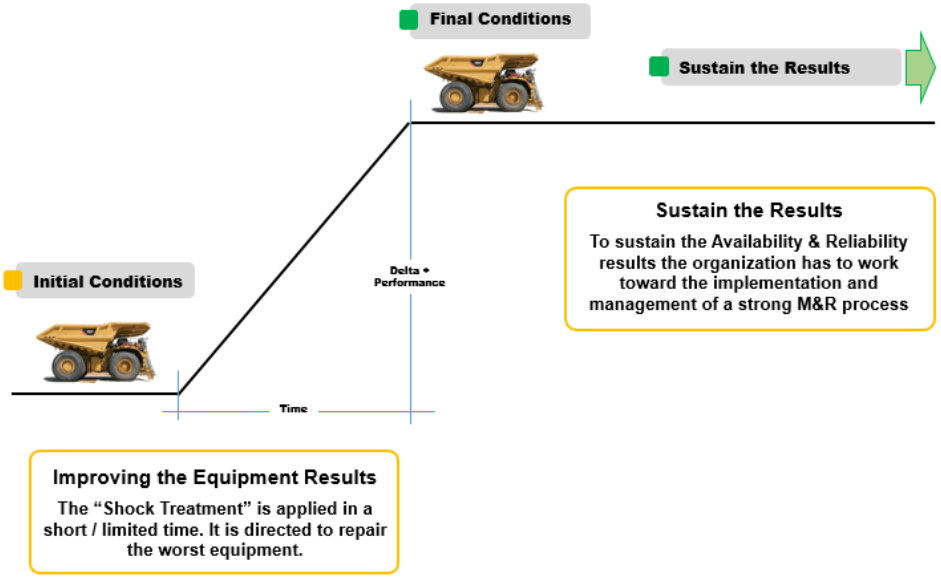
The procedure or action plan presented here is applicable whenever there might be a need to improve availability and reliability results of a mobile mining equipment fleet. This approach should be considered as a "shock treatment" and the objective should be to recover fleet performance in a short timeframe, not as a continuous maintenance practice or methodology.

The proposed recovery plan is based on selecting and repairing the worst performing machines, namely those that negatively influence overall fleet performance the most. By repairing the worst equipment, the final impact on fleet performance will be more significant, enabling performance goals to be achieved faster.

The following illustration graphically represents the recovery plan to improve fleet performance.

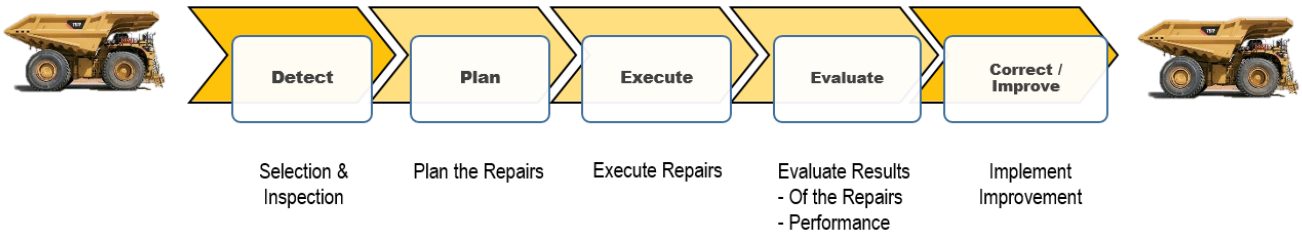


The recovery of equipment fleet performance must be supported by strong initiatives to improve or re-enforce the Maintenance & Repair Processes to "sustain the gains" and avoid performance deterioration to a level in where another recovery program must be applied.



The repair plan is designed to recover the "worst" perfomers and includes five different stages to follow. The recovery team includes Planning & Scheduling, Parts, and Service Departments.

The recovery process plan has 5 stages, abbreviated DPEEC.



Stage #1: Detection Phase.

There are two distinct activities performed in this phase:

1. The selection of equipment to enter the repair program.
2. Detailed machine inspections designed to determine the specific repairs / adjustments / upgrades needed to fix the causes of poor equipment performance.

Key points to consider in this Phase:

- Equipment selection: Although the Planning Department leads this, it's considered part of this stage to define and prioritize the worst performers.
- Performance analysis (for equipment selected): Thorough analysis is critical to prepare and focus inspection toward key areas or specific machine systems.
- Inspection: Conduct detailed machine inspections with the goal of determining all "remedies" (fixes) for equipment problems.
- Document corrective actions: Generate all backlogs and documentation to support the repair program definition.

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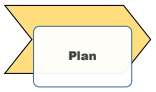


Stage #2: Planning Phase.

Planning drives three main activities:

1. Evaluation and selection of equipment to enter the repair program.
2. Definition of the detailed repair program.
3. Setup of targeted performance goals for equipment after the repairs.

Planning & Scheduling is involved in other stages, but now we will focus in stage 2.
Key points to consider in this stage:

- Equipment Ranking: Based on equipment performance, Planning determines equipment rankings used to select equipment that will enter the repair process.
- Lead the scheduling process for the detailed inspections and repairs.
- Determine the detailed repair plan based on the inspection results.
- Define the Post Repair performance targets of the equipment.

Stage #3: Execution Phase.

This stage is where all planned repairs completed. The main responsibilities of the repair crew are to perform with effectiveness (quality) and efficiency (speed), observing the duration and quality targets defined by Planning.
Key points to consider in this stage:

- Ensure completion of all repairs planned.
- Manage resources to achieve repair efficiency and quality.
- Maintain communication to the entire organization, informing of repair progress and new-found defects.
- Work closely with Planning to ensure the best program results.

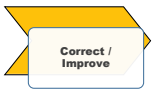
Stage #4: Evaluation Phase.

This stage includes two types of evaluation:

1. Execution of the repair program.
2. Performance of the equipment “post-repair”.

Key points to consider in this stage:

- Repair Program: What “went right” or what “went wrong” with respect to the program and the execution of it.
- Post-repair: Equipment performance results after returned to operation.
- Follow-up: It is very important to consider a follow crew to ensure machines operate as expected and symptoms of abnormal conditions are promptly fixed.
- It’s important to use the correct KPI´s and targets defined in the Planning Phase.



Stage #5: Correct / Improve Phase

Continuous Improvement is applied, directed to any phase of the process.
Key points to consider in this stage:

- The recovery team analyzes performance of each phase of the DPEEC process.
- Identify opportunities to improve.
- Generate 3W action plans to follow-up the corrective/improvement actions implementation.
- After validation of improvement actions, solutions must be incorporated into the recovery plan process.

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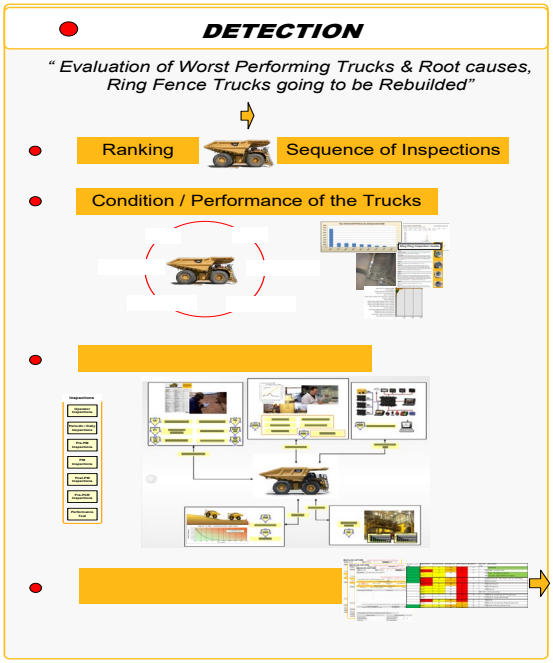
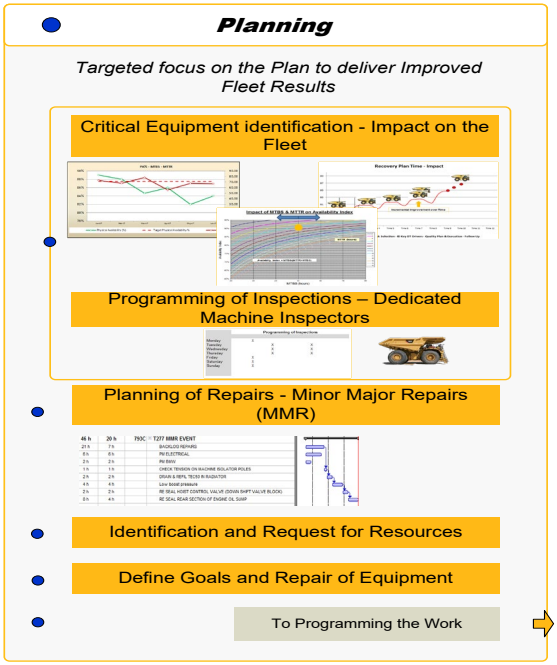
We recommend considering these five stages to execute a recovery plan.

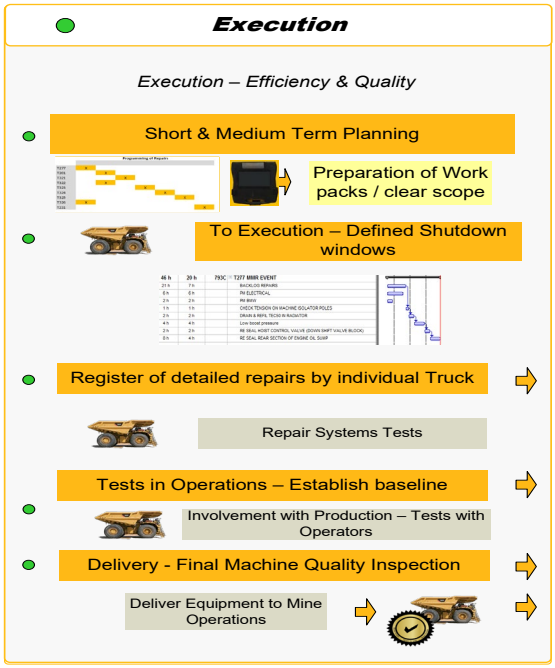


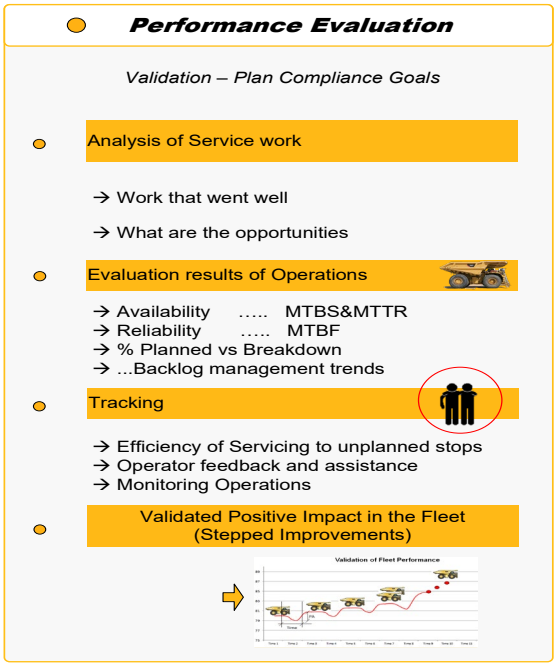


Plan Presented, Certified & Committed – Results % Scheduled

Detection

Planning





Monitor Performance (runtime) – MTBS Post repair – Availability Post repair – Effective use of Resources (MR)

Delivered Equipment Results – Confirmed & validated with Operations

Execution

Evaluation

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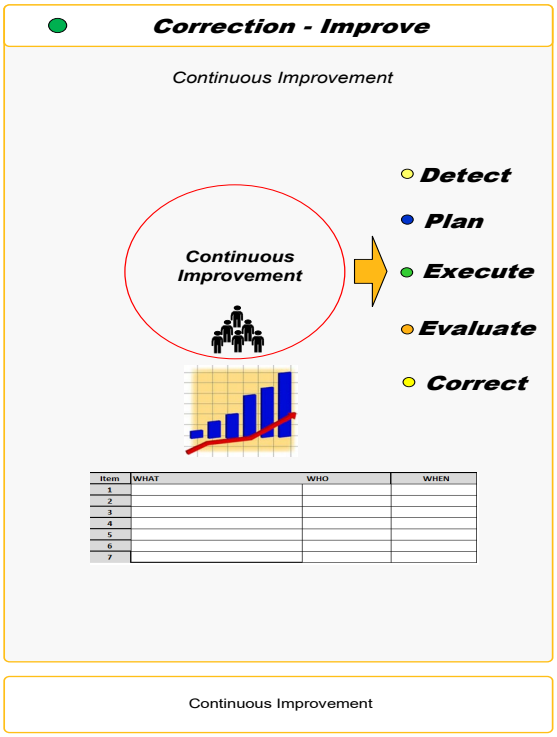

Correct - Improve

The preparation, resource mobilization, and implementation details are addressed in section 3.0.

3.0 Implementation Steps

This describes the steps to implement and execute this best practice, allowing the project to achieve successful results. Several adjustments introduced after the original plan are included here.

Customer's Needs Identification

The first step is to clearly identify the pain or customer needs, regarding equipment fleet performance. In the case of BVC, the concerns were 797F fleet availability and reliability results.

To identify the best course of action, a team of MATCO and Caterpillar specialists visited the mine and evaluated the maintenance and repair processes and practices. In this case, the customer provided all needed information and supported this evaluation.

This team highlighted several M&R opportunities that related to or caused low equipment performance. The team indicated the need to launch M&R process improvements, but also realized that fleet performance instability (reflected in high percent of unscheduled work) would make this effort difficult to successfully complete.

This challenge indicated that the best course of action would be to stabilize the fleet performance results (short-term) and improve M&R processes to ensure sustainability (medium/long term).

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Project Approval

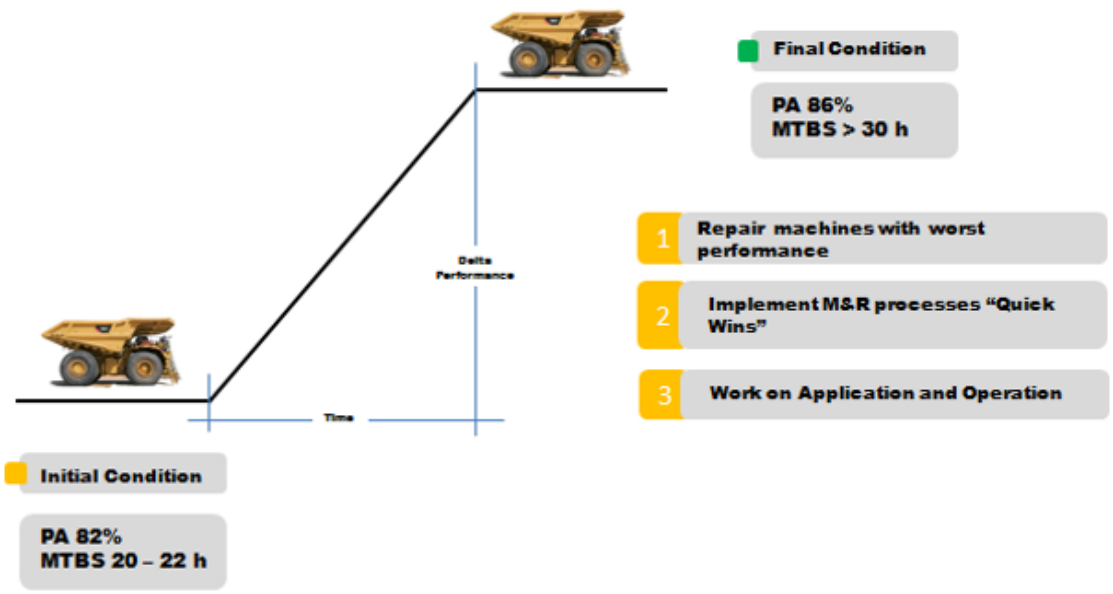
The MATCO and Caterpillar team presented a proposal to the customer mining department, which included two main initiatives, a short-term plan and medium/long term plan.

The short-term plan was designed as a “shock treatment”, to stabilize fleet performance in a short time period by identifying then repairing the worst performing machines. This would reduce workload of the M&R team, allowing them to focus on improving the M&R processes.

The medium/long term plan would focus to improve M&R processes, support the efforts of the maintenance organization, and allow them to be self-sufficient and achieve the expected equipment performance results.

In summary, the alternatives presented and accepted by Grupo Mexico, included a short-term project plan and a medium/long term project plan. The graphics below illustrate the objectives and the expected outcomes of each.

Short term Project



As mentioned before, short-term project objectives were to stabilize and improve the 797F fleet performance results. The three main activities considered were:

1. Repair the worst performing trucks.
    a. The main metrics to consider are physical availability and MTBS.
    b. Additional metrics for consolidation are MTBF and the ratio of MTBS/MTBF
2. Actively implement quick wins related to M&R processes.
3. Work to improve the equipment application and operation (if possible).

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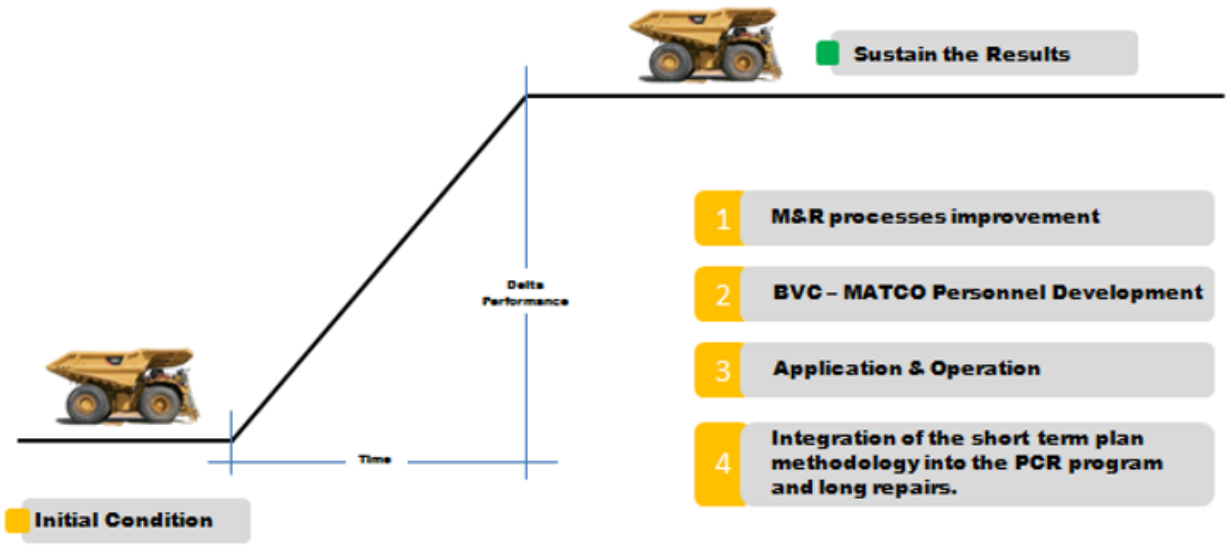

The main initiatives of the medium/long term plan were designed to provide the right tools and processes to the maintenance organization and enable M&R teams to sustain and manage the results achieved by the short-term plan. The four main initiatives were:

1. Improve and implement all updated M&R processes.
2. Develop processes owners of MATCO and BVC.
3. Continue working to improve and manage of the equipment application and operation.
4. Integrate the short-term plan methodology into the PCR program and long repairs.

Grupo Mexico agreed to move forward with the short-term plan, which is the main topic of this best practice. The team developed and documented a project charter to formalize the short-term plan. These objectives were to align the team, define the "pain", summarize the project objectives, and clearly document commitments and expectations.

Planning

The Planning department plays a critical role in structuring the plan, defining the timeline, and establishing expected performance results (in terms of availability). A big part of the recovery plan's success was to build a solid methodology to select machines and estimate the impact of each machine repaired.

The following is the calculation methodology used to determine the impact of each machine repaired, based on the initial availability performance of a specific machine. The impact calculation will easily highlight that highest positive impact on the equipment fleet is obtained by selecting and repairing the worst performing equipment. This methodology will also help determine the recovery time for availability results (from "X" availability to "Z" availability).

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Initial assumed conditions:

797F Truck Fleet	61
Calendar hours in a 30 days month	43920 h
Calendar hours per truck in a 30 days month	720 h
Initial Physical Availability	82%
Total Maintenance & Repair hours in a mont	7905.6 h

Table 1

Identify equipment for repair:

Identifying & prioritizing candidate trucks must be done carefully to ensure the correct forecasted impact of the repair programs. In this example, the candidate trucks were identified based on machine Physical Availability (PA). Machine reliability is another input to consider, but in this case, only physical available was used.

Worst Performing Trucks		Down Hours
Candidate 1	70%	216
Candidate 2	71%	209
Candidate 3	72%	202
Candidate 4	72%	202
Candidate 5	75%	180
Candidate 6	79%	151
Candidate 7	80%	144
Candidate 8	80%	144
Candidate 9	82%	130
Candidate 10	82%	130

Table 2

Hours Down = Machine Calendar hours – (Machine Calendar Hours x Machine PA %)

Define expected performance:

The next step is to define the expected PA goal after repairs are completed, or when the repaired truck is returned back in operation.

After Repair Performance Goal	
PA %	90%

Table 3

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With this target of 90% PA, the increment or impact on overall fleet availability can be calculated.

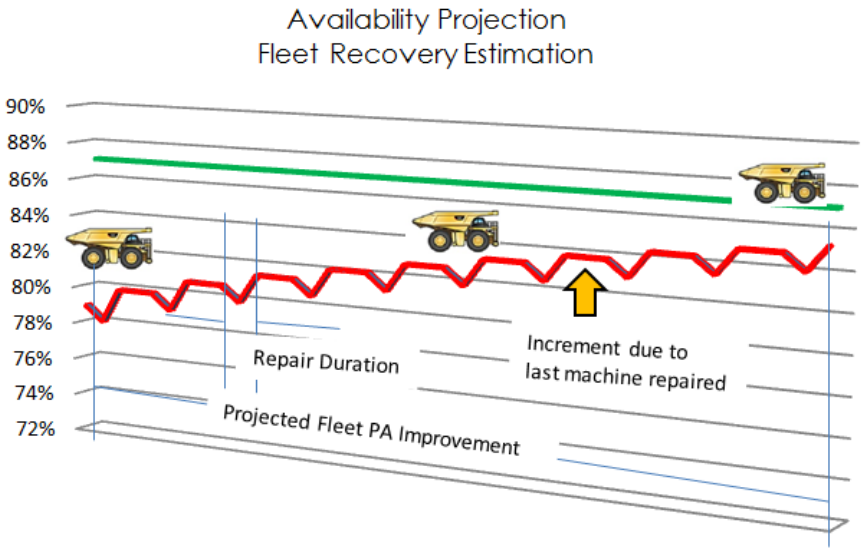
- Hours down at 90% PA = Machine Calendar Hours x (100% - PA target) [hours]
- Gain (DT hours) = Current Down Hours – Hours Down at 90 % PA [hours]

	Down Hours	Down Hours with 90% PA	Gain (Hours)	Estimated Fleet PA	PA Increment Cumulative	PA Increment per Machine
Candidate 1	216	72	144	82.3%	0.33%	0.33%
Candidate 2	209	72	137	82.6%	0.64%	0.31%
Candidate 3	202	72	130	82.9%	0.93%	0.30%
Candidate 4	202	72	130	83.2%	1.23%	0.30%
Candidate 5	180	72	108	83.5%	1.48%	0.25%
Candidate 6	151	72	79	83.7%	1.66%	0.18%
Candidate 7	144	72	72	83.8%	1.82%	0.16%
Candidate 8	144	72	72	84.0%	1.98%	0.16%
Candidate 9	130	72	58	84.1%	2.11%	0.13%
Candidate 10	130	72	58	84.2%	2.25%	0.13%

Table 4

Forecasted Fleet PA % = 100% – (Original Fleet M&R Hours – Gain) / Fleet Calendar Hours

The two columns on the right display the cumulative PA increment for the fleet and incremental PA per machine. With this projection, the total fleet recovery forecast can be represented graphically.



As it can be seen in the graph, each truck repaired will produce first a decrease in availability, due to the downtime related to each machine-specific repair program, and then an increase in availability due to the positive impact of returning a recovered truck to 90% physical availability.

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The positive increment by machine and of the entire fleet was calculated and presented in table 4. The decrease in availability due to the time the machine will be down for the repair program can be calculated as follows.

Repair Impact of	1 Truck / Month
Repair Program Duration	10 days
Machine Downtime	240 h
Impact on the Fleet PA	-0.55%

Table 5

Impact on fleet PA = # trucks under repair x repair duration (hours) / Fleet total calendar hours
(Another equation is "unavailability", namely: Impact% = Recovery DT hours / Fleet Total Calendar hours

By using this calculation methodology, the planning department can estimate the duration and the number of candidate machines required to meet the fleet availability goal. This is calculated by using actual machine performance data and estimated repair downtime hours.

In this specific project, it was calculated that to achieve target performance of 86% physical availability, the team needed to repair at least 25% of the 797 fleet, or 15 trucks. The calendar time required to repair 15+ trucks will depend on the number trucks repaired at the same time.

Once goals are defined for expected impact on machine and fleet performance, the results of the effort can be evaluated at any time and corrections can be applied if needed.

Equipment Selection

As mentioned previously, it is vitally important to correctly analyze equipment performance, select candidate machines, and prioritize equipment in the repair program.

The team discovered that careful attention should be paid to the selection process, since equipment performance results could be very unstable and variable. Due to this dynamic situation, the team needed a suitable system to easily read equipment performance and make correct decisions. For this reason, the team developed an equipment performance dashboard, shown in the next illustration.

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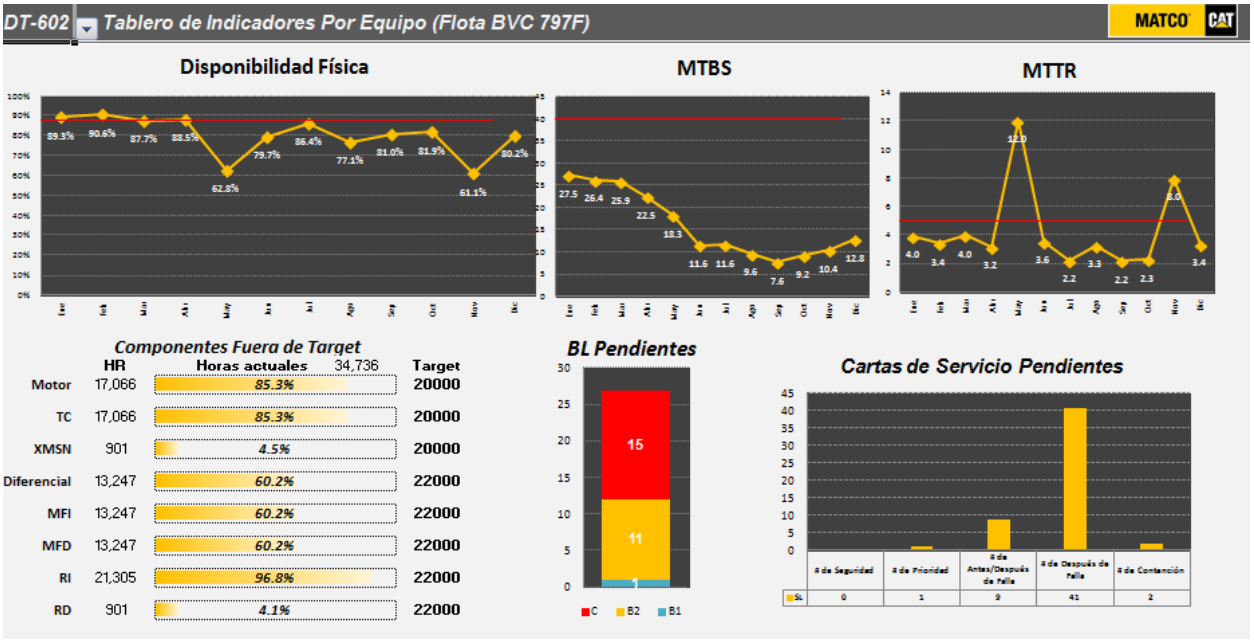

The information selected and included in the dashboard were:

- Physical Availability.
- MTBS.
- MTTR.
- Percentage of Component Usage.
- Pending Backlogs.
- Pending Service Letters.

The relationship between PA, MTBS, and MTTR must be carefully observed because availability can be directly influenced by performing short repairs (low MTTR). However, machine reliability, expressed as MTBS, cannot be bought as depicted in this example. Here additional metrics of MTBF and the ratio of MTBS/MTBF can also be considered.

Using this dashboard, the candidate list was built, with the objective to prioritize worst performers.

Resources

After defining the expectations and project goals, the team defined and mobilized the necessary resources. See section 5.0 for a detailed description of the resources allocated for this project.

Project start up and machine scheduling

At this stage, repair programs for specific machines could commence. In this situation, the ultimate decision to start the recovery plan was given by the customer's maintenance department. The steps this team needed to pay attention to were:

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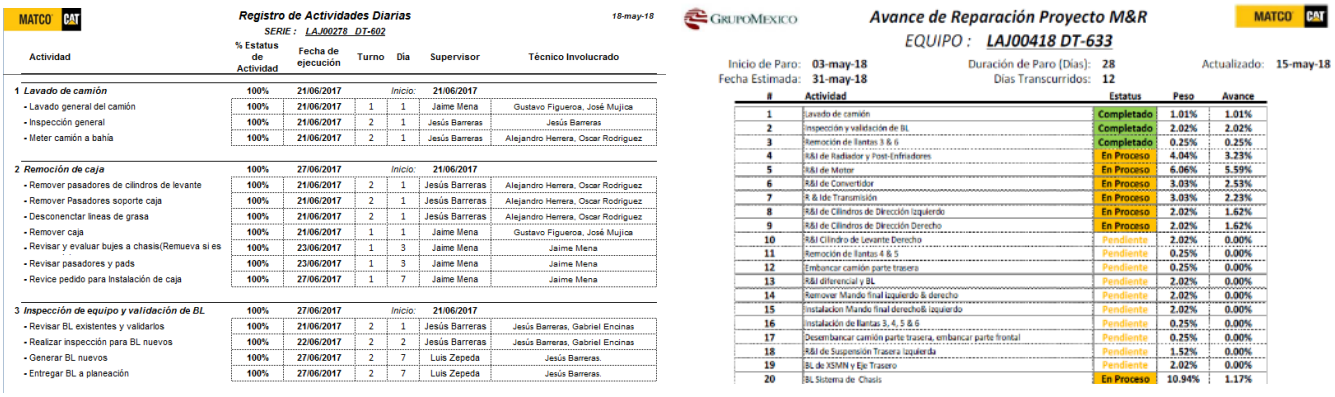
- Based on candidate selection, the first step was to schedule the in-depth machine inspection (the first stage of the recovery plan).
- Inspection results were processed by planning and scheduling, to build the repair program.
- A detailed repair program was presented to the customer for first truck candidate. The repair program included:
  - Equipment selection criteria.
  - List of activities, Gantt chart, and duration of the repair.
  - Parts and component list.
  - Quote for all required / recommended parts and components.
- Customer approval
- Work with customer planning and scheduling department to include the stoppage for the recovery truck into the weekly, short term program.
- Stop the machine and execute the repair program following the plan.
- It is important to remember that the five-step approach continues to select and define the repair program of the following truck candidates.

Follow Up and Control

Based on the team's experience, it's recommended to include the following activities in this phase:

- During execution of the repair program:
  - Daily execution progress gets validated.
  - Daily progress is promptly and effectively communicated to the organization.
  - The goals of prompt and effective communication are to:
    - Inform all parties.
    - Engage / involve the entire team to promptly and properly react to the unexpected new defects and difficulties found.
  - To formalize and communicate effectively, the team implemented two controls:
    - A one-page summary report
    - A technical bulletin board to be displayed in the repair bay.

The examples of the follow up report and bulletin board are shown below.



MATCO CAT

Registro de Actividades Diarias — 18-may-18

SERIE : LAJ00278 DT-602

Actividad	% Estatus de Actividad	Fecha de ejecución	Turno	Dia	Supervisor	Técnico Involucrado
1 *Lavado de camión*	100%	21/06/2017		*Inicio:*	21/06/2017	
- Lavado general del camión	100%	21/06/2017	1	1	Jaime Mena	Gustavo Figueroa, José Mujica
- Inspección general	100%	21/06/2017	2	1	Jesús Barreras	Jesús Barreras
- Meter camión a bahía	100%	21/06/2017	2	1	Jesús Barreras	Alejandro Herrera, Oscar Rodriguez
2 *Remoción de caja*	100%	27/06/2017		*Inicio:*	21/06/2017	
- Remover pasadores de cilindros de levante	100%	21/06/2017	2	1	Jesús Barreras	Alejandro Herrera, Oscar Rodriguez
- Remover Pasadores soporte caja	100%	21/06/2017	2	1	Jesús Barreras	Alejandro Herrera, Oscar Rodriguez
- Desconenctar lineas de grasa	100%	21/06/2017	2	1	Jesús Barreras	Alejandro Herrera, Oscar Rodriguez
- Remover caja	100%	21/06/2017	1	1	Jaime Mena	Gustavo Figueroa, José Mujica
- Revisar y evaluar bujes a chasis(Remueva si es	100%	23/06/2017	1	3	Jaime Mena	Jaime Mena
- Revisar pasadores y pads	100%	23/06/2017	1	3	Jaime Mena	Jaime Mena
- Revice pedido para instalación de caja	100%	27/06/2017	1	7	Jaime Mena	Jaime Mena
3 *Inspección de equipo y validación de BL*	100%	27/06/2017		*Inicio:*	21/06/2017	
- Revisar BL existentes y validarlos	100%	21/06/2017	2	1	Jesús Barreras	Jesús Barreras, Gabriel Encinas
- Realizar inspección para BL nuevos	100%	22/06/2017	2	2	Jesús Barreras	Jesús Barreras, Gabriel Encinas
- Generar BL nuevos	100%	27/06/2017	2	7	Luis Zepeda	Jesús Barreras
- Entregar BL a planeación	100%	27/06/2017	2	7	Luis Zepeda	Jesús Barreras

GRUPOMEXICO — *Avance de Reparación Proyecto M&R* — MATCO CAT

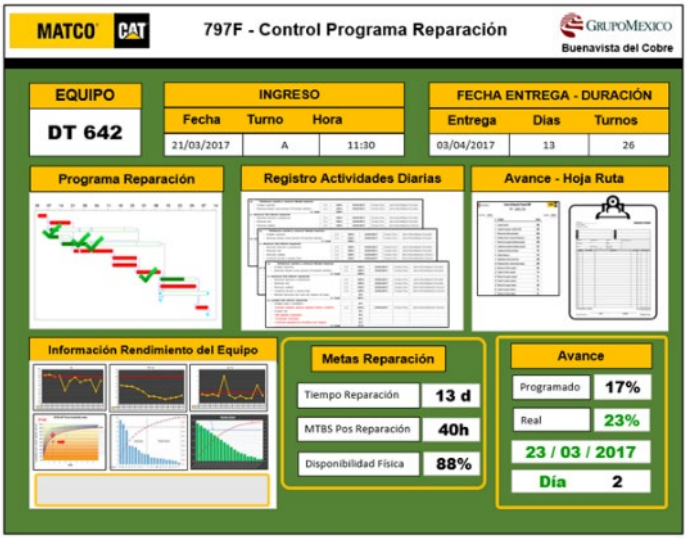
EQUIPO : LAJ00418 DT-633

Inicio de Paro: 03-may-18 — Duración de Paro (Dias): 28 — Actualizado: 15-may-18

Fecha Estimada: 31-may-18 — Dias Transcurridos: 12

#	Actividad	Estatus	Peso	Avance
1	Lavado de camión	Completado	1.01%	1.01%
2	Inspección y validación de BL	Completado	2.02%	2.02%
3	Remoción de llantas 3 & 6	Completado	0.25%	0.25%
4	R&I de Radiador y Post-Enfriadores	En Proceso	4.04%	3.23%
5	R&I de Meter	En Proceso	6.06%	5.59%
6	R&I de Convertidor	En Proceso	3.03%	2.53%
7	R & Ide Transmisión	En Proceso	3.03%	2.23%
8	R&I de Cilindros de Dirección Izquierdo	En Proceso	2.02%	1.62%
9	R&I de Cilindros de Dirección Derecho	En Proceso	2.02%	1.62%
10	R&I Cilindro de Levante Derecho	Pendiente	2.02%	0.00%
11	Remoción de llantas 4 & 5	Pendiente	0.25%	0.00%
12	Embancar camión parte trasera	Pendiente	0.25%	0.00%
13	R&I diferencial y BL	Pendiente	2.02%	0.00%
14	Remover Mando final Izquierdo & derecho	Pendiente	2.02%	0.00%
15	Instalacion Mando final derecho& izquierdo	Pendiente	2.02%	0.00%
16	Instalación de llantas 3, 4, 5 & 6	Pendiente	0.25%	0.00%
17	Desembancar camión parte trasera, embancar parte frontal	Pendiente	0.25%	0.00%
18	R&I de Suspensión Trasera Izquierda	Pendiente	1.52%	0.00%
19	BL de XSMN y Eje Trasero	Pendiente	2.02%	0.00%
20	BL Sistema de Chasis	En Proceso	10.94%	1.17%

One-page summary repair progress report



Bulletin Board Design and Implemented in the repair bay

This bulletin board was the method to easily display and effectively communicate the information to the shop organization. It was the "visual factory" for the project.

Equipment Delivery

The method and process to deliver the truck to production operations is another important step to consider. This is the opportunity for the maintenance & repair team to show the operations department what was targeted, what was accomplished, and let the operations department inspect and confirm all updates and repairs done (specifically targeting the truck operator and supervisor).

It is important to mention that in order to ensure the best possible truck operation, Grupo Mexico assigned fixed operators per truck returned to operation. This is vital to truck performance and facilitates the relationship with the maintenance team assigned to "follow up".

Photos of the first truck delivered to operation, truck 797F Internal Number 642

The operator receiving the truck and information of the repair details

Each truck repaired and returned to the operation received a distinctive identification.

This sign helps to quickly identify repaired machines and increase the pride and responsibility of the repair team.

Post Repair

After repairs are completed and machines returned to operation, the post repair "follow-up" routine starts. For this routine, the team included (3) distinctive tasks,

- "Post-Mortem" analysis performed right after the repairs are completed.
  - The objective is to document and analyze "what went right" and "what didn't go well" and "what needs to be improved".
- Follow up of the operational performance KPI's.
  - For this purpose, the team developed and used a special report that includes the top tier metrics such as, Physical Availability, Availability Index, MTBS, MTTR, % Scheduled Work.
- Corrections.
  - Corrections were applied based on the "post-mortem" analysis and the equipment performance results.

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Documentation of Benefits

As part of the implementation steps, it is important to include the closure and documentation of the benefits of the recovery plan. These benefits were documented, analyzed and validated in conjunction with the customer, Grupo Mexico.

The benefits were evaluated base on operational performance results of repaired machines and the return on investment (ROI). Details of benefits obtained are included in section 4.0.

4.0 Benefits

When analyzing the benefits of applying a recovery plan, it is important to consider the gains achieved in six different areas.

- Equipment performance
- Production and operation
- Profit, return of investment
- Maintenance and repair workload
- Dealer support – Trusted partner

Equipment Performance

The first documented benefit after repair completion was operational performance. The following are the performance results of the first three 797F trucks repaired, namely Availability, Utilization, Availability index, MTBS, MTTR, and MTBF. These trucks have customer id #642, #600 & #602.

797F Truck #1: #642

The first truck that was selected and repaired had the following performance indicators.

Performance Metrics (KPI's) of Truck 642

Performance KPI	Before Repair		After Repair					
	Dec 2016- Mar 2017		July 2017		January 2018		Accum to Apr 2018	
Physical Availability	77.9	%	94.9	%	94.6	%	91.2	%
Utilization of Availability	71.9	%	93	%	91.4	%	91.3	%
Availability Index	74.9	%	94.6	%	94	%	90.4	%
MTBS	18.9	h	43.8	h	37.9	h	39.4	h
MTTR	6.4	h	2.5	h	2.4	h	4.2	h
MTBF		h	65.7	h	58.5	h	56	h

The results shown below highlight the trend and time to stabilize the final performance.

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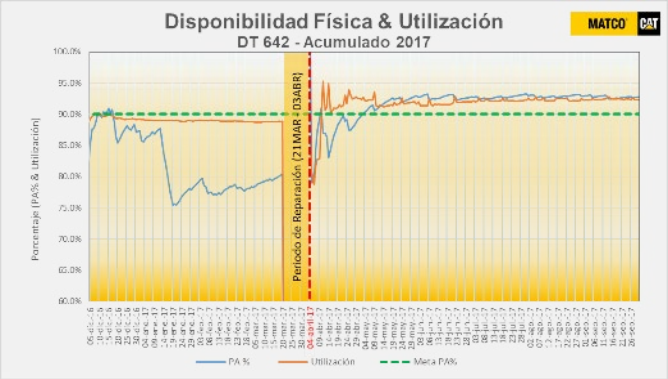
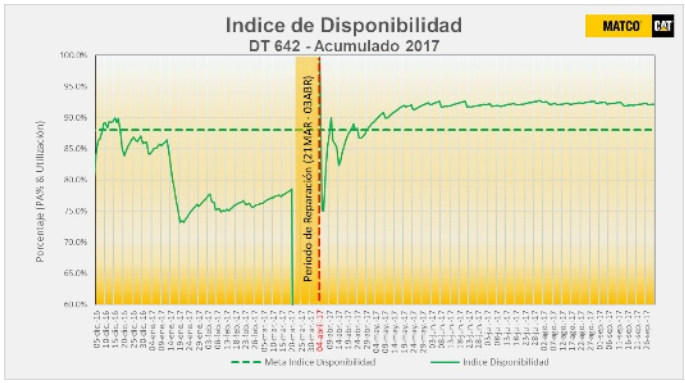
Physical Availability, Utilization and Availability Index

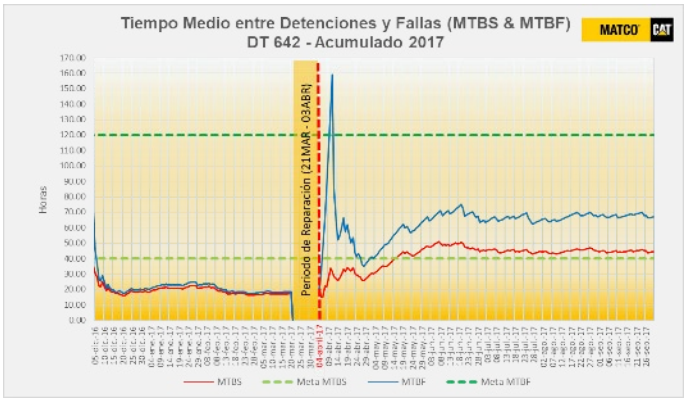


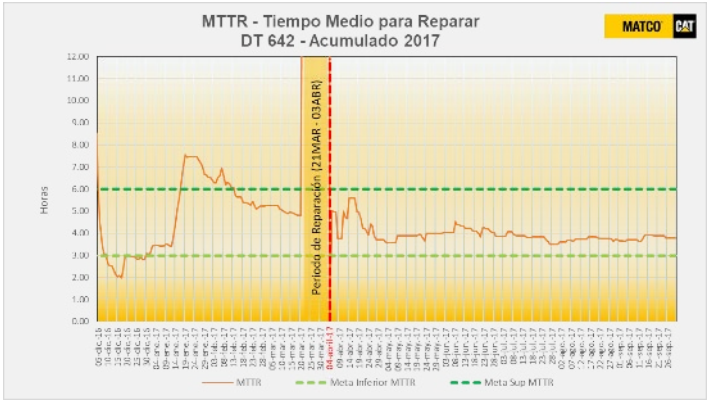


As shown in the Availability, Utilization and Availability Index graphs, after the initial period to stabilize results, the machines reached stable performance results above 90%.

It is important to highlight that utilization also increased, indicating that the production department was satisfied with conditions and performance truck. Production likes reliable equipment.

MTBS, MTBF and MTTR





Production and Operation

In the short term, the production department received tangible results that directly influenced truck utilization and operator satisfaction. Each truck candidate included a complete repair and upgrade of the operator station, which also included installing a commercial radio in the cab, which was very well received by the operator.

The combination of equipment performance results, improved operator station, and operator comfort support the increased utilization and subsequent contribution to production goals.

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Return of Investment – ROI

MATCO and Grupo Mexico defined the return on investment (ROI) for the first trucks repaired. The table below includes the benefits regarding availability, operating hours, and production (tonnes).

	DT642	DT600	DT602
Availability Gained	14.90%	11.50%	15.30%
Operating Hours Increase	108.8	83.8	111.7
Incremental in Tonnage	47,414.20	33,544.50	48,690.00

With the incremental operating hours and increased production related for repaired trucks, the ROI was calculated between 1 to 4 months. See table below for more details.

	DT642	DT600	DT602
Return of Investment wo Components (Months)	1.06	2.11	1.93
Return of Investment with Components (Months)	1.51	4.12	3.68

As shown in the ROI results table, the team differentiated repairs including components and without including components. This is calculated because the repair stoppage was also used as a window of opportunity to replace components included in the PCR program.

Maintenance and Repair Workload

Another benefit to highlight is related to the decrease in M&R workload of the maintenance organization. Post-repair trucks that returned to operation performed with higher availability and reliability, which translated into fewer scheduled and unscheduled maintenance stoppages.

The maintenance organization benefits from decreased workload and can allocate more time to improve M&R processes. This is important to sustain the equipment performance results.

Dealer Support – Trusted Partnership

This recovery plan presented a valuable opportunity to both dealer and customer teams. Both organizations joined efforts toward the common goal of effectively supporting the production plan of BVC operation. The time working together, addressing challenges, and successfully implementing and executing the recovery plan was very positive to recognize strengths, respect each other, and build trust. This built a positive foundation for future challenges.

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5.0 Resources Required

Successfully implementing a fleet recovery plan such as this one, requires assigning adequate resources to ensure the proper execution of all steps and phases of the plan. It's recommended for the recovery plan to be implemented separately from the regular daily M&R activities. That means that the resources here included are to be considered 100% dedicated solely to the recovery plan.

When properly selected and managed, the five types of resources noted below guarantee the established goals for future recovery repairs will be achieve without setbacks. The main types are:

- ✓ Human Resources – Personnel
- ✓ Manual and specialized tools
- ✓ Facilities – Repair shop bay and Offices
- ✓ Parts and Components
- ✓ Vehicles – Follow up and Service

HR – Personnel

Personnel is possibly the most important resource to be considered and selected in advance of project startup. The project team followed and recommend are to carefully select adequate people to fill the responsibilities of:

- Leadership
- Repair Execution

The people assigned to leadership roles were tasked with the following responsibilities:

- Customer relationships.
- Project management.
- Planning.
- Coordination of parts and components.
- Guide and manage the execution of repairs.
- Supervise the "follow up" of equipment already repaired.

The technicians selected for repair execution must have:

- Solid knowledge of the machines being repaired,
- Experience in component replacement (remove, install, commissioning),
- Troubleshooting,
- Data analysis,
- Inspections routines,
- A very proactive approach to failure avoidance
- Full commitment for all safety guidance and procedures.

The quantity of technicians and supervisors will depend on the quantity of machines to be repaired in parallel (simultaneously). The quality and experience of technicians is critical, and these resources cannot be underestimated. In the case of this project, the team selected and recruited the best technicians throughout the entire MATCO dealership.

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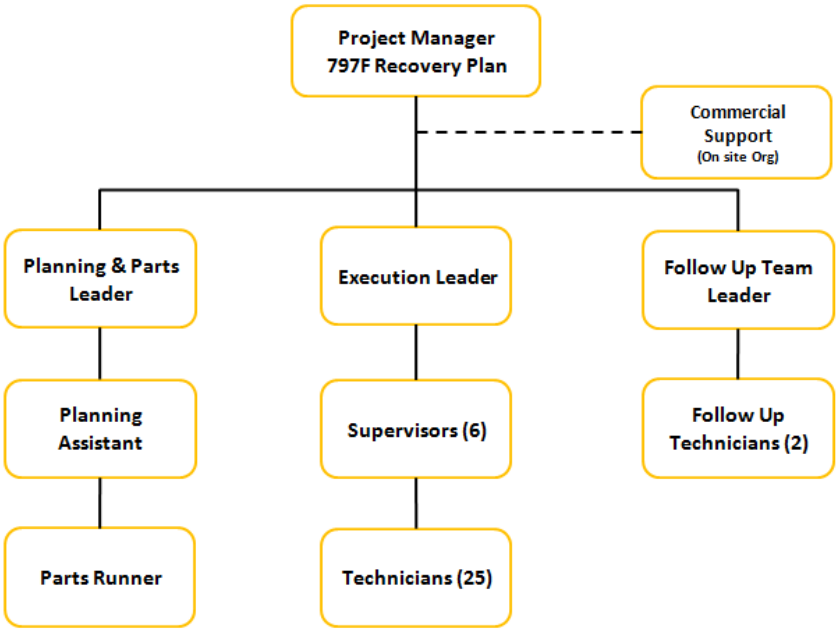

It is important for all team members to receive a “project induction”, in the key project elements, overall goals and objectives, importance to the company, and their role and responsibilities in meeting the expected results.

For this project, the management team allocated adequate time to perform this induction for all team members that arrived on site to be integrated in the project. It was clearly communicated to all dealer personnel that they were NOT called to fix machines but were selected to be part of a very important project for the customer and dealership.

Manual and Specialized Tools

The objective is to have the right tool at the right time for the right work. Tools are a critical element to ensure efficiency and effectiveness of repairs executed by the technician teams.

The leaders and supervisors for repair execution selected and acquired the tools required for the technicians. The attached tools list does not include common tools that all technicians have in their toolboxes.

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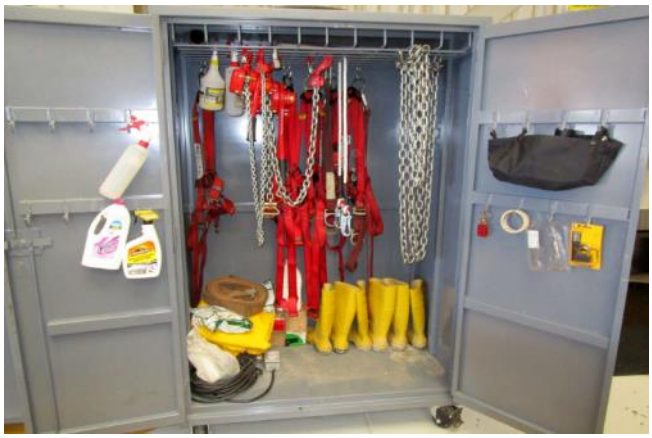
As a summary the following table illustrates the investment in tools for a project.

Type of Tools	Qty	Cost USD
Pneumatic	27	$ 565.22
Inspection	16	$ 410.85
Torque - Manual	6	$ 1,095.06
Hytorc - Hydraulic	15	$ 51,064.59
Diagnostic	12	$ 6,218.10
Air Conditioning	9	$ 725.48
Engine - Intake	16	$ 632.99
Engine - Exhaust	3	$ 1,687.12
Hoist - Lifting devices	32	$ 3,501.49
Miscellaneous	3	$ 3,781.24
		$ 65,900.89

The organization and handling of tools is equally important as the tools themselves and it's vital to ensure the proper conditions and availability of tools when required. The team designed and allocated suitable cabinets and installed in the assigned shop repair bay. See pictures below.



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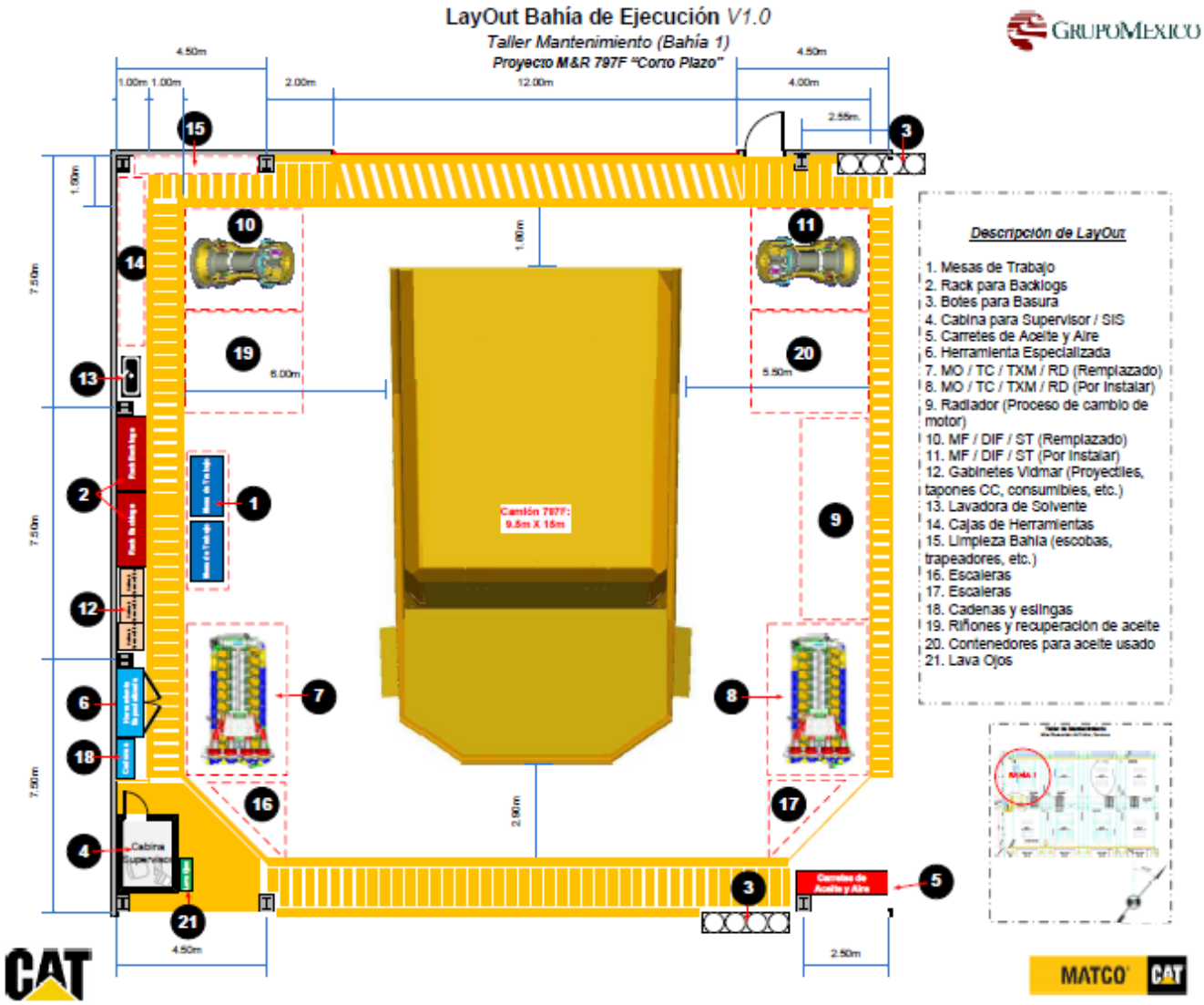
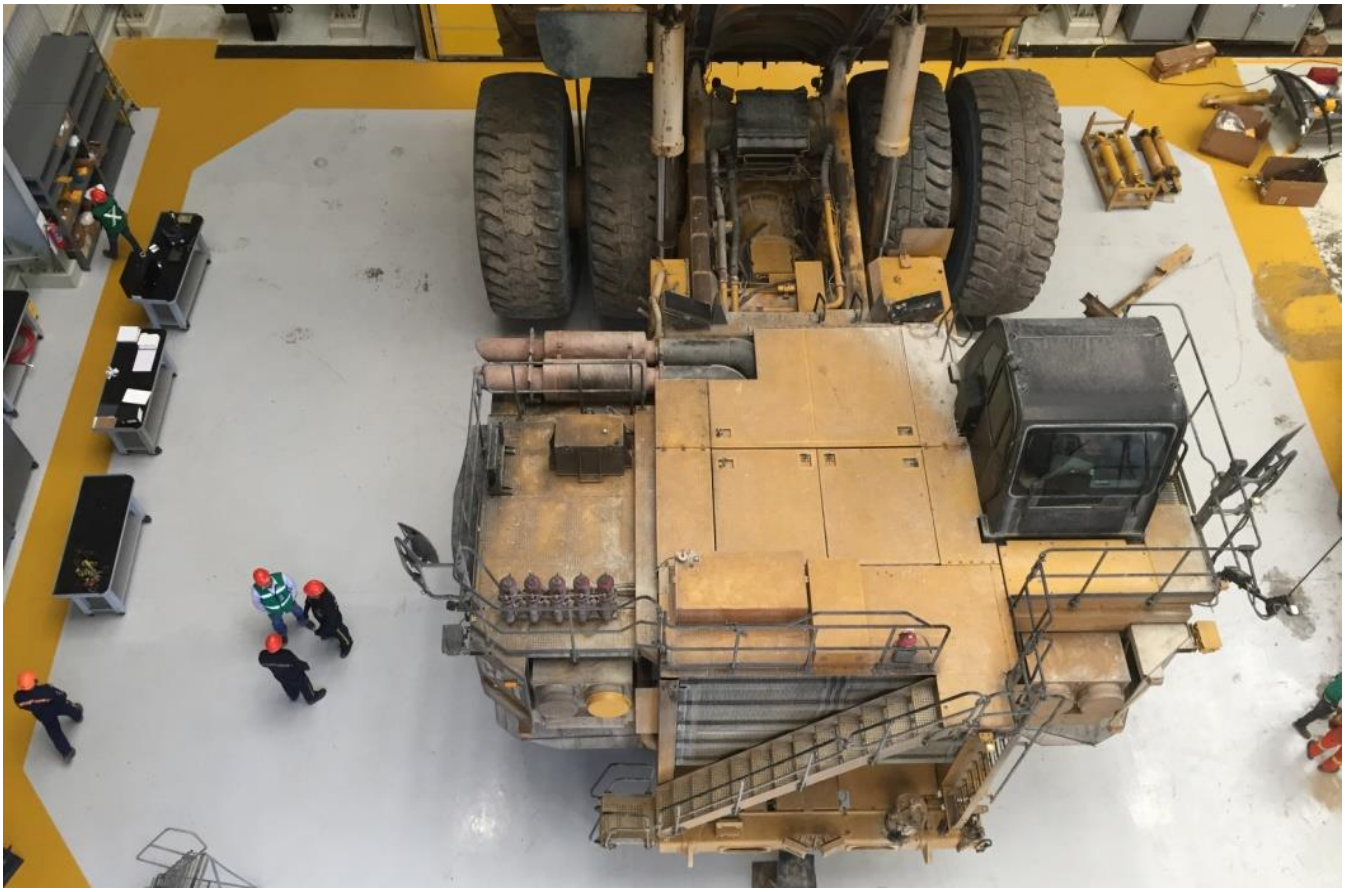
Facilities – Repair shop bay and Offices

Another critical resource to consider is facilities, including the repair shop bay and office space. It is important to remember that the recovery project be performed in parallel to all other regular M&R activities. For that reason, it is important to assign a shop bay and office space for the project team.

The customer, Grupo Mexico, understood the importance of allocating project resources and they assigned a shop bay exclusively for the recovery project. Below is the layout schematic used to equip the shop bay to meet the requirements of all targeted repair activities.





A mobile office was added, specially designed to accommodate all the functions of project management and planning & scheduling. The following images show the office design and set up.

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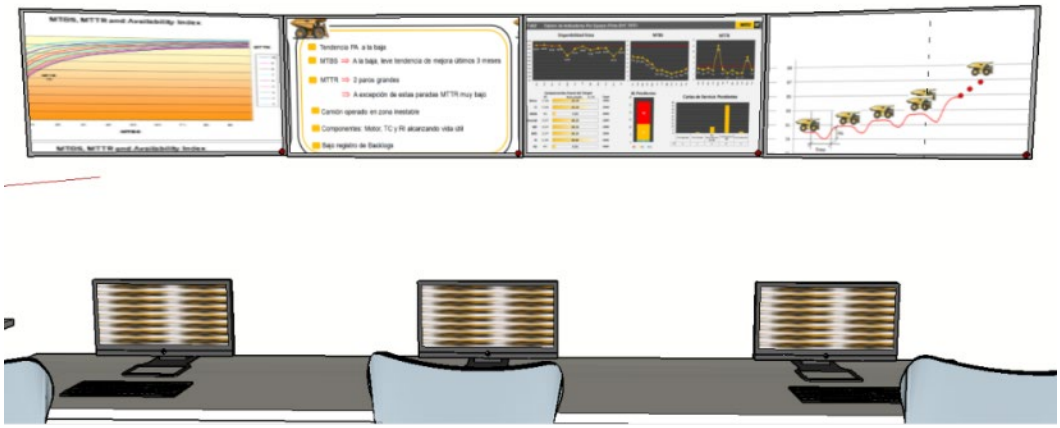

The investment to prepare the shop bay and office space was approximately $43,000 USD.

Shop bay and Office space Investment (USD)		
Shop Bay	$	15,528.77
Office	$	27,871.25
	$	**43,400.02**

Parts and Components

After completion of the inspection phase, the planning department identify all repair activities to be completed, with the correspondant Parts and Components (P&C). It's important not to understimate the time and effort required to acquire parts and component. This includes Quotation, Availability and Parts orders, Authorization, Parts delivery and Storage in the shop repair bay.

The quantity and consideration of parts and components will be driven by the conditions of your equipment fleet. Regarding components, the percent of remaining life and component health are variables that will define whether to include or not include component exchanges during the program.

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Vehicles – Follow-up and Service

The team included a "follow-up stage" to ensure the best possible results of the recovery plan. "Follow-up" includes analysis of performance KPI's and physically following repaired trucks in operation. The main functions and actions of the follow-up team were:

- Quick response to unscheduled stops.
- Field inspections.
- PM inspections.
- Backlogs generation and execution.
- Analysis and proper reaction to performance KPI's.
- Direct support to machine operators.
- Application and operation advisory.

Project leadership must provide all necessary resources to facilitate and assure correct execution of this important function. This mainly consisted of a dedicated service vehicle, equipped with all needed regular hand and diagnostic tools.

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6.0 Supporting Attachments / References

1. Project Charter
2. Equipment and tooling list (hand tools, workshop and offices).
3. Metrics Report

BVC Project Charter - M&R (Corto Plazo).pc

BVC Metrics report.pdf

BVC Tooling (Shop Inspection Office).xlsx

7.0 Related Best Practices

None

8.0 Acknowledgements

Special recognition should be given to the managers of Grupo Mexico's Cananea operation, whom supported the implementation and successful results of the recovery plan presented here.

This best practice was written by:

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Mission

Improve Fleet Results for the 793 Haul Trucks at Cananea – Grupo Mexico to effectively support Production and prepare the Fleet for Implementation of Maintenance and Repair Processes

PROJECT DEFINITION

STRATEGIC PILLARS

Pillar #1 – SPECIALIZED REPAIR TAILORED TO EACH TRUCKS PROCESSED

Pillar #2 – REPAIRS DONE IN PARALLEL WITH THE REST OF SHOP ACTIVITIES

Pillar #3 – EACH PHASE IS DETAILED DEFINED AND MEASUREABLE

Pillar #4 – RESULTS ARE MEASURED & CONTROLED BASED ON THE REPAIRED UNITS

Pillar #5 – SPECIALIZED RESOURCES ASSIGNED TO THE PROJECT

Pillar #6 – INTEGRATION WITH THE REST OF THE ON GOING IMPROVEMENT INITIATIVES

Pillar # 7 – INTEGRATION OF THE MAINTENANCE AND PRODUCTION DEPARTMENT

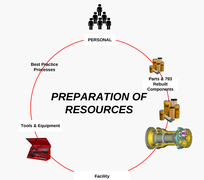
Pillar #8 – INTEGRATION OF GRUPO MEXICO – MATCO - CATERPILLAR

MATCO CAT

GRUPOMEXICO

CAT

PREPARATION



Timing of inspections – Generation of Backlogs – Detection levels

DETECTION

" Evaluation of Worst Performing Trucks & Root causes, Ring Fence Trucks going to be Rebuilded"

Ranking → Sequence of Inspections

Condition / Performance of the Trucks

Execution of Inspections

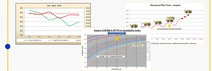
Definition of Repair Actions Needed

Plan Presented, Certified & Committed – Results % Scheduled

Planning

Targeted focus on the Plan to deliver Improved Fleet Results

Critical Equipment identification - Impact on the Fleet



Programming of Inspections – Dedicated Machine Inspectors

Planning of Repairs - Minor Major Repairs (MMR)

Identification and Request for Resources

Define Goals and Repair of Equipment

To Programming the Work

Monitor Performance (runtime) – MTBS Post repair – Availability Post repair – Effective use of Resources (MR)

Execution

Execution – Efficiency & Quality

Short & Medium Term Planning

Preparation of Work packs / clear scope

To Execution – Defined Shutdown windows

Register of detailed repairs by individual Truck

Repair Systems Tests

Tests in Operations – Establish baseline

Involvement with Production – Tests with Operators

Delivery - Final Machine Quality Inspection

Deliver Equipment to Mine Operations

Delivered Equipment Results – Confirmed & validated with Operations

Performance Evaluation

Validation – Plan Compliance Goals

Analysis of Service work

→ Work that went well

→ What are the opportunities

Evaluation results of Operations

→ Availability ….. MTBS&MTTR
→ Reliability ….. MTBF
→ % Planned vs Breakdown
→ ...Backlog management trends

Tracking

→ Efficiency of Servicing to unplanned stops
→ Operator feedback and assistance
→ Monitoring Operations

Validated Positive Impact in the Fleet (Stepped Improvements)

Continuous Improvement

Correction - Improve

Continuous Improvement

- Detect
- Plan
- Execute
- Evaluate
- Correct

Items	WHAT	WHO	WHEN
1			
2			
3			
4			
5			
6			
7			

Mining Equipment Management

Cananea- Grupo Mexico Project

2017

MATCO CAT

CAT

Maintenance & Repair (M&R) Implementation

Planteamiento de Proyecto

Nombre del Proyecto: **Plan Corto Plazo Flota 797F**

Situación Deseada
(¿Qué beneficios se obtendrán? / ¿Qué visión tengo?)

La producción actual y futura de la operación BVC, depende en gran medida de las horas máquina disponible (disponibilidad de equipo) y su correspondiente confiabilidad (mínima interrupciones de proceso).

Nuestra visión de la situación futura de la flota de camiones 797F,es que estos operen en un ambiente controlado, satisfaciendo las metas de disponibilidad y confiabilidad a un óptimo costo de mantenimiento. Esto se consigue a través del trabajo en conjunto de mantenimiento y producción, aplicando procesos y técnicas de gestión de vanguardia de administración de equipos móviles de minería (MEM).

Para **conseguir la estabilidad** en los resultados de la flota, debemos asegurar un mínimo estandar de condición en los equipos. Esto nos proporciona una base para una implementación efectiva de gestión de MEM.

Finalmente la estabilidad del mantenimiento y los resultados de la flota, entregan al departamento de producción una flota de equipo confiable con la que puede trabajar **para obtener la óptima productividad** de los equipos.

Oportunidad
(¿Cuál es la situación actual?)

- La actual gestión de mantenimiento y producción están siendo desafiadas por los resultados de la flota de camiones 797F

- El nivel y variabilidad de los resultados en terminos de disponibilidad, confiabilidad y eficiencia de las reparaciones (PA 82%, MTBS 20h, MTTR 4-5h), sobrecargan las tareas de mantenimiento y a la vez producen interrupciones que afectan negativamente el cumplimiento de las metas de producción.

- Procesos de mantenimiento con variadas oportunidades.

- Recursos desafiados para emprender una recuperación de flota al mismo tiempo de mantener funcionando las tareas de mantenimiento habitual.

Objetivos
(¿Cuáles son los objetivos del proyecto?

Y = f (X)

Y= Incrementar la disponibilidad física a 85% de la flota de camiones 797F de BVC

X1= Incrementar la disponibilidad física de cada camión intervenido a 88%
X2= Intervenir XX camiones en el período de Marzo a Diciembre del 2017
X3= Enfoque del plan de recuperación de la flota en las máquinas más problemáticas (bandidos)
X4= Ejecutar el plan de recuperación de la flota en forma paralela a la operación de mantenimiento.
X5= Plan de recuperación soportado por una efectiva estructura de trabajo, infraestructura, herramienta, refacciones, etc.
X6= Diseño y ejecución de plan de recuperación siguiendo el concepto de linea de producción con etapas muy definidas (DPEEC).
X7= La recuperación de los KPI's de la flota se basan en indicadores específicos de las máquinas problemáticas.
X8= Efectivo control y seguimiento de los camiones intervenidos (Follow-up)
X9= Fuerte involucramiento de departamento de producción (Operación).

Alcance
(¿Cuáles son las limitantes del proyecto?)

Alcance de proyecto:

- Principal: Flota de camiones 797F de Buenavista del Cobre (BVC).
- Secundario: Contribuir al desarrollo de personal y procesos.

Fuera de proyecto:
Este proyecto no tiene responsabilidad directa sobre:
- Implementación de proyectos de mejora de procesos de mantenimiento y reparación (M&R) de BVC.
- Proyecto formal de desarrollo de personal de mantenimiento de Matco y BVC.

Plan del Proyecto
(¿Qué actividades se van a seguir? ¿Cuándo se van hacer?)

- **Inicio:** 16 / Enero / 2017
- **Fin:** XX / Diciembre / 2017

1. → Definir estrategia de recuperación de flota (Enero 2017)

2. → Definir recursos en cantidad y tiempo (Enero 2017)

3. → Aprobación / Involucramiento de niveles gerenciales Grupo México y Matco (Enero 2017), Caterpillar (Febrero 2017)

4. → Movilización y preparación de recursos, preparación de logística para arranque de proyecto. Instalación de recursos / lugar de trabajo (Febrero / Marzo 2017)

5. → Planificación de secuencia de reparaciones e impacto en resultados. (Febrero 2017)

6. → Ejecución del plan aplicado al primer equipo (DPEEC) (Abril 2017)

7. → Diseño e implementación de reportes de retroalimentación / reporte de resultados / correción de desviaciones (Marzo 2017)

8. → Gestión de proyecto - ejecución y seguimiento de puntos de revisión de avance (Mensual)

9. → Cumplimiento de metas (Entregables), traspaso de experiencias y cierre de proyecto (Diciembre 2017)

Selección del Equipo
(¿Integrantes? ¿Cuáles son sus roles?)

- **Dueños de Proceso:**

- **Patrocinador:**
  Alfredo Fonseca, 1 Grupo México, Josh Meyer

- **Líderes de Proyecto:**
  Jesús Campos, 1 Grupo México (Recomendación: 1 persona de departamento de planeación)

- **Equipo de Implementación:**
  - Planificación: Martín F. Barredez
  - Programación: XXXX
  - Líder de Ejecución: Miguel Olivas
  - Equipo de Técnicos:

- **Responsible / Coordinator:**
  Terry L. Miller (Site Performance Manager)
  José Gutiérrez (Dealer Mining Coordinator)
  1 Grupo Mexico (BVC)

- **Soporte SME:**
  Carlos Luque (Matco)
  Abelardo Flores (MEM)
  Erik Tautkus (PPM)
  Grupo Cameron McGovern (MEM)

- **Black Belt:**
  Glenn R. Carrasco (Matco)

BUENAVISTA DEL COBRE - PROYECTO RECUPERACION FLOTA 797F - M&R CORTO PLAZO

MATCO CAT

SEGUIMIENTO DE RESULTADOS EQUIPOS RE-ACONDICIONADOS

RESUMEN EJECUTIVO

EQUIPO REPARADO	FECHA ENTREGA	RESULTADOS AL DÍA	SMU ENTREGA
DT 642	abril 3, 2017	septiembre 20, 2017	**25,786 horas**

RESUMEN EJECUTIVO

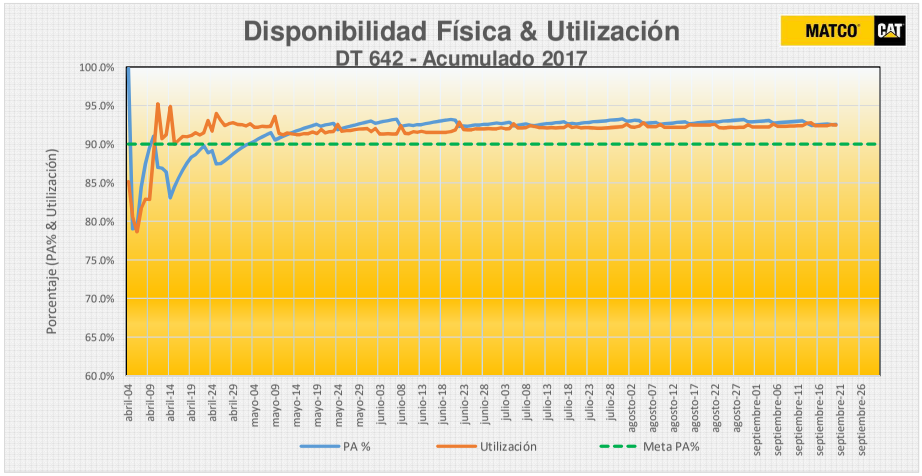
El seguimiento incluído en este reporte comienza a partir de las 7:00 AM del día 4 de Abril.

RESULTADO MENSUAL MÉTRICAS DE PRIMER NIVEL (TOP TIER)

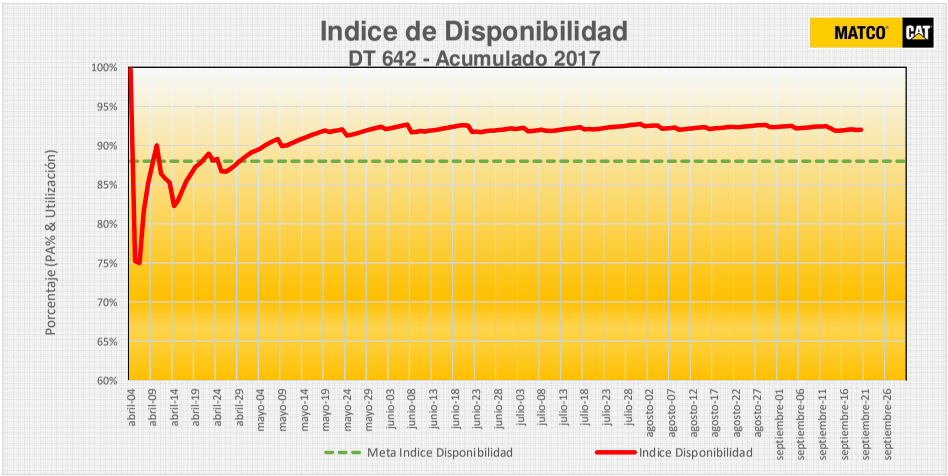
KPI - Métrica	Antes Rep. (Dic 16 a Feb 17)	META		ACUM. septiembre 20, 2017	ABRIL	MAYO	JUNIO	JULIO	AGOSTO	SEPT. septiembre 20, 2017
Horas Operadas				**3494**	556.6	654.6	610.8	657.2	**626.8**	**410.0**
Disponibilidad Física	77.9 %	90%		**92.6%**	89.5%	96.1%	92.3%	94.9%	**91.5%**	**90.3%**
Utilización Disponibilidad	71.9 %	85%		**92.5%**	92.5%	91.6%	91.9%	93.0%	**92.1%**	**94.6%**
Index Disponibilidad	74.7 %	88%		**92.0%**	88.8%	95.7%	91.7%	94.6%	**90.8%**	**89.8%**
MTBS	18.9 h	40	h	**44.8**	29.3	109.1	40.7	43.8	**44.8**	**45.6**
MTTR	6.4 h	3 - 7	h	**3.9**	3.7	4.9	3.7	2.5	**4.5**	**5.2**
MTBF		120	h	**68.5**	39.8	163.7	61.1	65.7	**78.3**	**82.0**

GRÁFICAS DE TENDENCIAS (ACUMULADO 2017)

EQUIPO REPARADO	FECHA ENTREGA	FECHA DE REPORTE	SMU ENTREGA
DT 642	abril 3, 2017	septiembre 20, 2017	25,786 horas



Disponibilidad Física y Utilización se encuentran estables por arriba de la meta.



Índice de Disponibilidad se muestra estable por arriba de la meta

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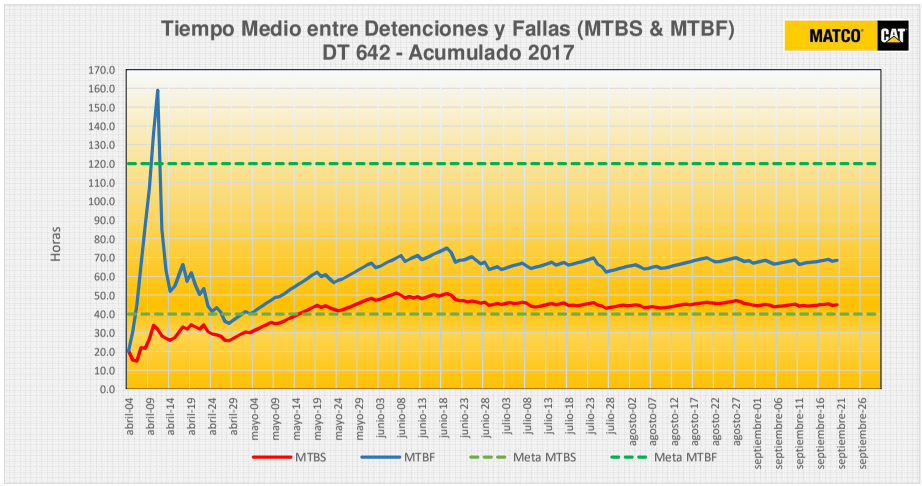
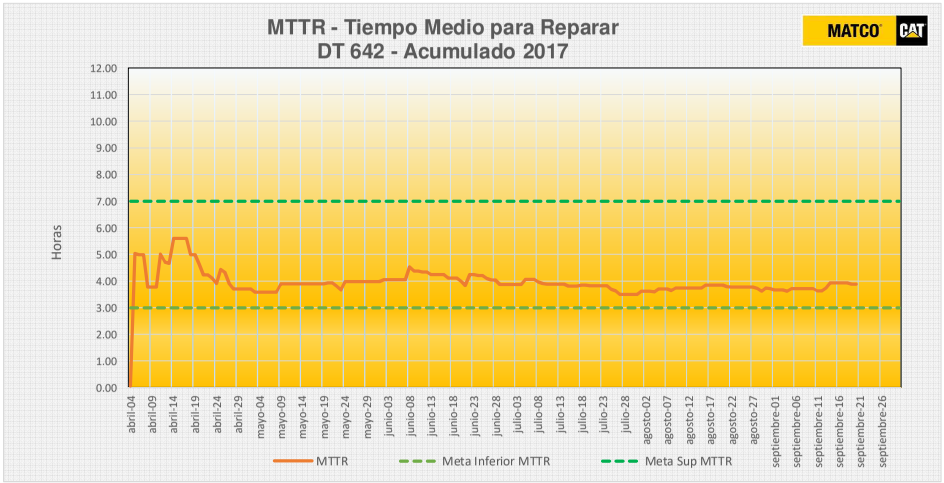


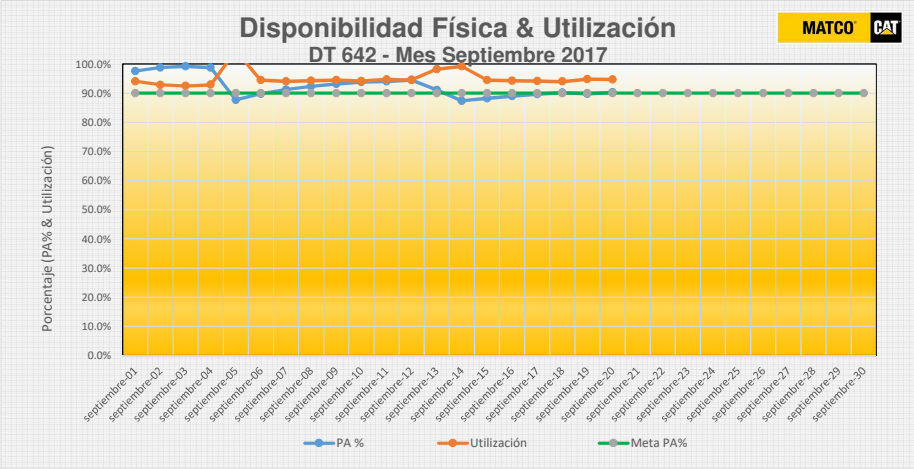
La gráfica muestra que el MTBS y MTBF están estables.



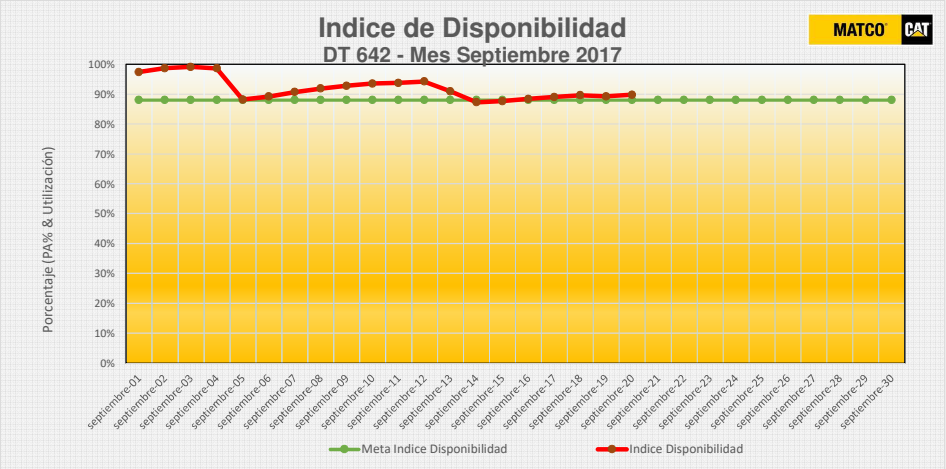
MTTR se ha mantenido estable

GRÁFICAS DE TENDENCIAS (SEPTIEMBRE 2017)

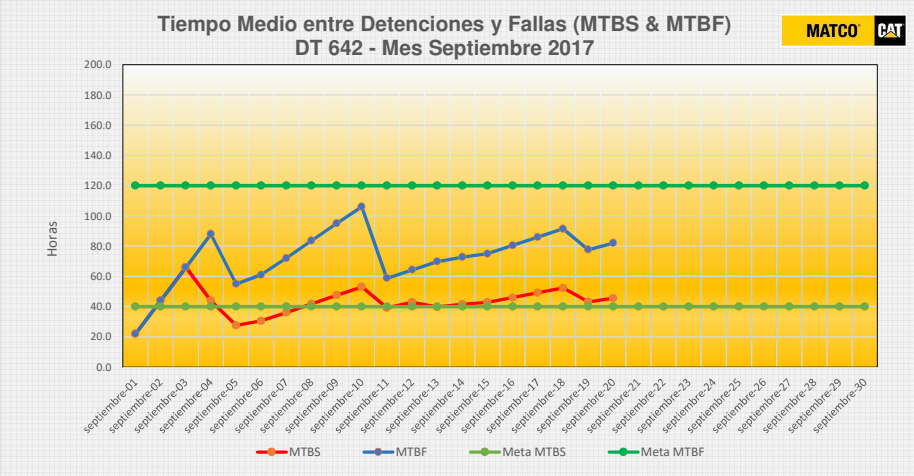
EQUIPO REPARADO	FECHA ENTREGA	FECHA DE REPORTE	SMU ENTREGA
DT 642	abril 3, 2017	septiembre 20, 2017	25,786 horas



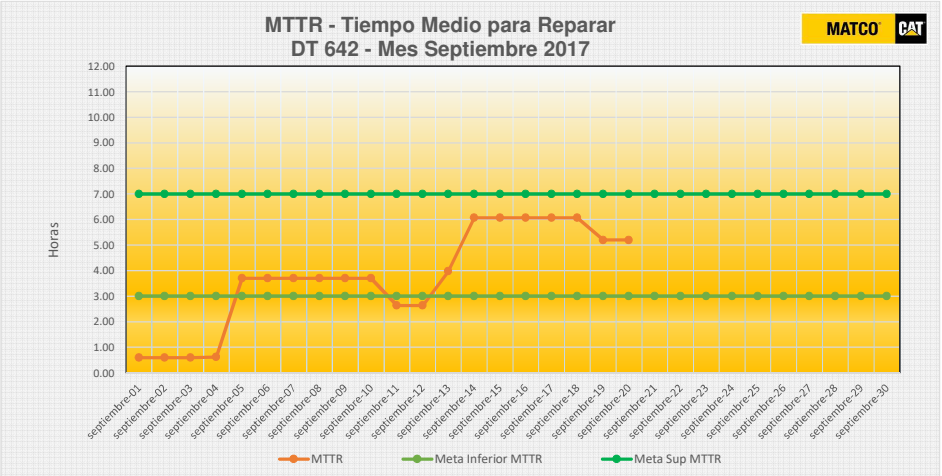
Disponibilidad Física



Índice de Disponibilidad.



MTBS y MTBF.



MTTR dentro de los límites

LISTADO DE HERRAMIENTAS PARA EJECUCIÓN

PROYECTO "M&R CORTO PLAZO 797F BVC"

#	No. Parte	Descripción	Cantidad	Precio Unit.	Total	T.M	Proveedor	T. Entrega
Herramienta Neumática								
1	222-3048	PISTOLA DE IMPACTO 3/8	2					
2	222-3050	PISTOLA DE IMPACTO 1/2	4					
3	222-3053	PISTOLA DE IMPACTO 3/4	2					
4	222-3054	PISTOLA DE IMPACTO 1	1					
5	222-3055	PISTOLA DE IMPACTO 1	1					
6	222-3065	PULIDORA GRANDE	3					
7	5P-9718	RESPALDO PULIDORA	10					
8	222-3074	BURIL	2					
9	223-1159	LAPIZ NEUMATICO	2					
Herramienta de Torque / Ajuste								
1	9U-5020	TORQUE 3/8	2					
2	9U-5006	TORQUE 1/2	2					
3	9U-5007	TORQUE 3/4	1					
4	QC3ETT250	PROBADOR DE TORQUE SNAP-ON	1					
Herramienta de Torque / HYTORC (Cambio de componentes)								
1	AVANTY ICE-3	HYTORC 1" (460-3084 FT-LBS)	1					
2	AVANTY ICE-5	HYTORC 1" (804-5360 FT-LBS9	1					
3	STEALTH-4	LLAVE (604-4020 FT-LBS)	1					
4	STEALTH-4	CARTUCHO 55 MM	0					
5	STEALTH-4	CARTUCHO 46 MM	1					
6	STEALTH-2	LLAVE (278-1869 FT-LBS)	1					
7	STEALTH-2	CARTUCHO 46 MM	1					
8	JGUN-A3-SP	PISTOLA NEUMATICA DE 1"	2					
9	JGUN-A1-SP	PISTOLA NEUMATICA DE 3/4"	1					
10	JGUN-FRL-A	REGULADOR DE PRESIÓN	1					
11	AV3-HEAD	BRAZO DE REACCIÓN P/SUSP. DELANTERAS	1					
12	020046MM	DADO IMPACTO 1" X 46 MM	1					
13	020055MM	DADO IMPACTO 1" X 55 MM	1					
14	080055MM	DADO IMPACTO 1 1/2" X 55 MM	1					
15	HY-115-2	FUENTE DE PODER PARA AVANTY Y STEALTH	1					
Herramienta Especializada (2da. Parte)								
3		**INSTALACIÓN DE BELLOWS**						
	396-9127	HERRAMIENTA BELLOWS MANIFULES	1					
	421-4079	HERRAMIENTA BELLOWS SALIDA TURBINA	1					
	421-4080	HERRAMIENTA BELLOWS ENTRADA TURBINA	1					
4	RD51858	ASPIRADORA 16 GL USO RUDO						
5	176-6536	EXTRACTOR (TRANSMISIÓN)	2					
6	5P-8715	EXTRACTOR (SUSP. TRASERAS)	6					
7		KIT DE MUESTREO COMPLETO	1					
8		**PRESIÓN DE ACUMULADORES**						
	175-5507	Nitrogen Charging Group	1					
	9U-6740	Accumulator Charging Gp	1					
	152-2023	Nitrogen Charging Fitting Gp	1					
	334-1071	Adapter	1					
	9U-6741	Adapter	1					
Herramienta de Levante								
1	439-3938	ESLABONES 3/8	4					
2	439-3939	ESLABONES 3/4	4					
3	439-3940	ESLABONES 1/2	4					
4	439-3941	ESLABONES 1 1/4	3					
5	439-3942	ESLABONES 7/8	3					
6	439-3943	ESLABONES 1 1/2	3					
7	1U-8263	BANDA 4.88 M 4218 KG	1					
8	1U-8252	BANDA 3.66 M 2812 KG	1					
9	1U-8246	BANDA 2.44 M 2109 KG	1					
10	1U-8245	BANDA 4.27 M 1406 KG	1					
11		PULPOS 3 GANCHOS 11,200 LBS	1					
12		PULPOS 4 GANCHOS 39,000 LBS	1					
13		CADENAS CON GANCHOS 4 MTS	2					
14	URR5886	MANERAL DE 1"	1					
15	URR47015	TECLE 750 KG PALANCA	1					
16	URR47315	TECLE 3.2 T PALANCA	1					

TOTAL	MXN	$	-
	USD	$	-

LISTADO DE HERRAMIENTAS PARA INSPECCIÓN

PROYECTO "M&R CORTO PLAZO 797F BVC"

#	No. Parte	Descripción	Cantidad	Precio Unit.	Total	T.M	Proveedor	T. Entrega
Herramienta de Inspección								
1	FABRICACIÓN	EXTENCIÓN ELECTRICA 15 MTS USO RUDO	3					
2	4C-6478	LAMPARA TRABAJO 15.24 M	2					
3	1U-7321	LAMPARA MANO	3					
4	PECE55	CEPILLOS DE ALAMBRE	5					
5	1U-9366	CINTA METRICA	2					
6	441-5254	PISTOLA PARA LAVAR CON SOLVENTE	1					
Herramienta de Diagnóstico								
1	317-7484	COMUNICADOR (Cambia a 466-6258)	2					
2	305-5528	CABLE VIMS 3G	2					
3	127-9797	VIMS STD	1					
4	257-9140	MULTIMETRO	2					
5	198-4240	INDICADOR DE PRESIÓN DIGITAL	1					
6	349-4200	TERMOMETRO DIGITAL	1					
7	6V-7830	TETRAGAUGE	2					
8	6V-4157	INDICADORES DE PRESIÓN (TRANSMISIÓNES	1					
Herramienta Especializada								
1		KIT COMPLETO PARA A/C						
	208-1356	BOMBA DE VACIO A/C	1					
	208-1341	MANOMETROS R134	1					
	208-1335	MANGUERA AZUL	1					
	208-1336	MANGUERA ROJA	1					
	272-3091	MANGUERA AMARILLA	1					
	208-1339	COPLE BAJA	1					
	208-1340	COPLE ALTA	1					
	6V-7145	CASE	1					
	9U-5325	TERMOMETRO	1					
2		HERRAMIENTA PRUEBA ADMISIÓN	1					
	321-6022	PRESURIZADOR	2					
	4C-8858	COPLE	2					
	030-8370	ADAPTADOR	2					
	4C-7046	REGULADOR	1					
	4C-7091	VALVULA	1					
	4C-7120	CONECTOR	2					
	8L-6005	TEE	1					
	3B-7722	CONECTOR	1					
	123-5176	MANGUERA	1					
	032-3506	ADAPTADOR	1					
	3B-7255	CONECTOR	1					
	4C-7159	CONECTOR	1					
3		KIT DE MUESTREO COMPLETO	1					
Escaleras								
1	BYPETA26	ESCALERA DE 2 MTS	2					
2	BYPETA33	ESCALERA DE 1 MTS	2					

TOTAL	MXN	$	-
	USD	$	-

Lista de Equipamiento de Oficina

PROYECTO "M&R CORTO PLAZO 797F BVC"

Cant.	Descripción	Precio Unit.	Monto	Moneda	Ubicación
Equipamiento de Computo					
2	Computadoras de Escritorio				Área de planeación / Taller.
2	Pantallas de 42"				4 en planeación.
1	Escaner ScanJet Pro 2500 f1				Área de planeación.
1	Impresora a color				Área de planeación.
2	Laptos con ET - Vims - SIS				Taller.
2	Camaras fotográficas				Inspección / Ejecución
1	Internet para Oficina				Oficina
1	Banda Ancha Móvil (Backup)				Oficina
	Total:				
Mobiliario					
1	Oficina Móvil 12x60				
1	Acondicionar Oficina				
1	Muebles Oficina				
	Total:				
Equipamiento Oficina					
3	Pizarrones de corcho				En las 3 áreas.
3	Pintarrones				En las 3 áreas.
1	Botiquín de primeros auxilios.				1 área
2	Radios con frecuencia de mina				Taller.
4	Radios con frecuencia interna				Taller / Oficina.
Equipamiento Eléctrico y Red					
6	Conexiones 110 V				Sala de juntas.
8	Conexiones 110 V				Sala de Juntas (Centro de mesa)
1	Conexiones HDMI				Sala de Juntas (Centro de mesa)
1	Conexión telefónica				Sala de Juntas (Centro de mesa)
2	Conexión Ethernet				Sala de Juntas (Centro de mesa)
6	Conexiones 110 V				Oficina de Ejecución
2	Conexión Ethernet				Oficina de Ejecución
14	Conexiones 110 V				Área de planeación.
5	Conexión Ethernet				Área de planeación.

Lista de Equipamiento de Bahías

PROYECTO "M&R CORTO PLAZO 797F BVC"

BAHÍAS Y CONTENDOR DE HERRAMIENTAS

MARCA	MODELO	CANT.	DESCRIPCIÓN	ID LAYOUT	PRECIO	MONEDA	SUBTOTAL	COMENTARIOS
VIDMAR	VIDMAR	2	GABINETE VIDMAR PARA CONTENEDOR DE HERRAMIENTAS / SUMINISTROS	12				KIT DE LIMPIEZA DE MANGUERAS Y CONSUMUBLES
ULINE	H-3916	2	RACK DE REFACCIONES (Estantería de Acero para Uso Pesado - 72 x 36 x 84")	2				PARA COLOCAR KITS DE BACKLOGS CLASIFICADOS POR SISTEMA
CAT	4C-3875	1	LAVADORA DE SOLVENTE (121.92 x 55.88 x 25.40 cm)	13				EN ESPERA DE FECHA TENTETIVA DE LLEGADA
IMM	FABRICAR	1	GABINETE PARA HERRAMIENTA ESPECIALIZADA	6				RESGUERDAR HERRAMIENTA DEL PROYECTO
IMM	FABRICAR	1	GABINETE PARA CADENAS Y ESLINGAS	18				SOLO PARA EQUIPO DE LEVANTE
IMM	FABRICAR	4	MESAS DE TRABAJO (MOVILES)	1				
IMM	FABRICAR	1	CASETA DE SUPERVISOR	4				
IMM	FABRICAR	4	CANASTAS PARA PARTES EN WIP					CANASTAS PARA RESGUARDO DE PARTES A REUTILIZAR DURANTE LA REPARACIÓN
IMM	FABRICAR	8	BASE PARA CAJA DE HERRAMIENTAS	14				
BRADLEY	Lava ojos: # Grainger: 4YF98 # Fabricante: S19-921 Base Lava ojos: # Grainger: 3NY85 # Fabricante: S19-399	2	LAVA OJOS (Está disponible en la página de Grainger en dos secciones)	21				VER CON CLIENTE
	FABRICAR	2	ESCALERAS	16				Ver con Cliente
	FABRICAR	2	ESCALERAS CHICAS	17				Ver con Cliente
		1	Riñon de Aceite Eje Trasero	19				Ver con Cliente
		1	Riñon de Aceite de Transmisión	19				Ver con Cliente
		1	Riñon de Aceite de Dirección	19				Ver con Cliente
		1	Riñon de Aceite de Hidráulico	19				Ver con Cliente

TOTAL	MXN	$	-
	USD	$	-